

An all-cuts PPT-entangled tripartite Werner state

Autonomous proof candidate for specialist verification

9 August 2026

Abstract

We answer affirmatively the open question in Remark 5.5 of Park–Jung–Park–Youn: for every $d \geq 3$, there is a $U \otimes U \otimes U$ -invariant state on $(\mathbb{C}^d)^{\otimes 3}$ which is positive under partial transpose and entangled across each of the three bipartitions. In the coordinates of Eggeling–Werner, the example is the rational point

$$(r_+, r_-, r_1, r_2, r_3) = \left(\frac{3}{4}, \frac{1}{32}, 0, 0, \frac{3}{16} \right).$$

It is invariant under the cyclic permutation of the three systems. The one-cut PPT criteria hold strictly, whereas the corresponding biseparability criterion fails by 45/1024. A bounded search found no later explicit resolution, so novelty remains subject to specialist review.

1 Source question

Park, Jung, Park, and Youn [1] construct tripartite Werner states which are PPT-entangled across two cuts but not PPT across the third. In Remark 5.5 on PDF page 25 they ask for a state which is PPT-entangled across all three cuts. The complete source passage is reproduced below.

$$V_{(12)}\rho_t V_{(12)} = \frac{1}{d^3 + (t+1)d^2 + 2t} \left(\frac{u+t}{d} V_e + V_{(23)} + \frac{t}{d} V_{(123)} + \frac{t}{d} V_{(132)} \right) \quad (5.19)$$

does not satisfy the CCP condition (5.10).

It might be interesting if we can find a tripartite PPT-entangled Werner state with respect to all the three partitions A - BC , B - AC , and C - AB . However, Lemma 7 of [EW01] implies that there is no such an example $\rho = \sum_{\sigma} a_{\sigma} V_{\sigma}$ if one of the following conditions is satisfied:

- $a_{(12)} = a_{(13)}$ and $a_{(123)} = a_{(132)}$,
- $a_{(13)} = a_{(23)}$ and $a_{(123)} = a_{(132)}$,
- $a_{(23)} = a_{(12)}$ and $a_{(123)} = a_{(132)}$,

We leave the general situation as an open question.

In the source's terminology, a tripartite Werner state is a density operator ρ on $(\mathbb{C}^d)^{\otimes 3}$ satisfying

$$(U \otimes U \otimes U)\rho(U \otimes U \otimes U)^* = \rho \quad (U \in \mathcal{U}_d).$$

It is, for example, A - BC PPT if $\rho^{T_A} \geq 0$, and it is A - BC entangled if it is not separable across that bipartition.

2 Eggeling–Werner coordinates

We recall only the exact criteria used below from Eggeling–Werner [2]. For $\sigma \in S_3$, let V_σ permute the three tensor factors, and put

$$\begin{aligned} R_+ &= \frac{1}{6}(\text{Id} + V_{(12)} + V_{(23)} + V_{(13)} + V_{(123)} + V_{(132)}), \\ R_- &= \frac{1}{6}(\text{Id} - V_{(12)} - V_{(23)} - V_{(13)} + V_{(123)} + V_{(132)}), \\ R_0 &= \frac{1}{3}(2\text{Id} - V_{(123)} - V_{(132)}), \\ R_1 &= \frac{1}{3}(2V_{(23)} - V_{(13)} - V_{(12)}), \\ R_2 &= \frac{1}{\sqrt{3}}(V_{(12)} - V_{(13)}), \quad R_3 = \frac{i}{\sqrt{3}}(V_{(123)} - V_{(132)}). \end{aligned}$$

For a Werner state define $r_k = \text{Tr}(\rho R_k)$. Such a state is characterized by

$$r_+, r_-, r_0 \geq 0, \quad r_+ + r_- + r_0 = 1, \quad r_1^2 + r_2^2 + r_3^2 \leq r_0^2. \quad (1)$$

For the cut A – BC , Lemma 10 of [2] says that PPT is equivalent to the state conditions together with

$$\begin{aligned} 0 &\leq r_-, & 0 &\leq 1 + r_1 - r_+ - r_-, \\ 0 &\leq 1 - r_1 - 5r_- - r_+, & 0 &\leq -1 - r_1 + r_- + 5r_+, \\ r_2^2 + r_3^2 &\leq \frac{(1 - r_1 - 5r_- - r_+)(-1 - r_1 + r_- + 5r_+)}{3}, \\ r_2^2 + r_3^2 &\leq (1 - r_1 - r_- - r_+)(1 + r_1 - r_- - r_+). \end{aligned} \quad (2)$$

We also need the negative branch of the biseparability criterion in Theorem 5 of [2]. If $0 \leq r_- < 1/3$ and

$$-1 < u := \frac{1 + r_1 - r_- - 2r_+}{1 - 3r_-} \leq 0, \quad (3)$$

then A – BC separability requires

$$3r_2^2 + 3r_3^2 + (1 + 2r_1 + r_- - r_+)^2 \leq (2 + r_1 - 4r_- - 2r_+)^2. \quad (4)$$

Together with the other branch and three isolated extreme points, the cited theorem makes these conditions necessary and sufficient. Our point lies strictly inside the parameter range (3), so only (4) is relevant.

3 Proof intuition

The obstruction quoted in the source applies when a transposition symmetry is present. We instead impose only cyclic symmetry. In the Eggeling–Werner Bloch coordinates, setting $r_1 = r_2 = 0$ makes the state invariant under the three-cycle, so the three bipartitions have identical PPT and separability status. A nonzero r_3 breaks the transposition symmetries. It remains only to choose rational r_+, r_-, r_3 inside the PPT body but outside the biseparable body; the point below does so with strict rational margins.

4 Construction and proof

Theorem 1. *For every integer $d \geq 3$, define*

$$\begin{aligned} \rho_d = & \frac{9}{2d(d+1)(d+2)}R_+ + \frac{3}{16d(d-1)(d-2)}R_- \\ & + \frac{21}{64d(d^2-1)}R_0 + \frac{9}{32d(d^2-1)}R_3. \end{aligned} \quad (5)$$

Then ρ_d is a tripartite Werner state which is PPT and entangled across each of $A-BC$, $B-AC$, and $C-AB$.

Proof. The trace dimensions of the three blocks give

$$(r_+, r_-, r_0, r_1, r_2, r_3) = \left(\frac{3}{4}, \frac{1}{32}, \frac{7}{32}, 0, 0, \frac{3}{16} \right). \quad (6)$$

Indeed, the coefficient-to-expectation relations are

$$r_+ = \frac{d(d+1)(d+2)}{6}c_+, \quad r_- = \frac{d(d-1)(d-2)}{6}c_-, \quad r_i = \frac{2d(d^2-1)}{3}c_i \quad (0 \leq i \leq 3).$$

Thus $\text{Tr } \rho_d = 1$. Condition (1) holds because

$$r_3^2 = \frac{36}{1024} < \frac{49}{1024} = r_0^2, \quad (7)$$

so $\rho_d \geq 0$. Formula (5) is a linear combination of permutation operators and therefore is $U \otimes U \otimes U$ invariant.

We first examine the cut $A-BC$. At (6), the nontrivial linear quantities in (2) are

$$1 - r_+ - r_- = \frac{7}{32}, \quad 1 - 5r_- - r_+ = \frac{3}{32}, \quad -1 + r_- + 5r_+ = \frac{89}{32}. \quad (8)$$

Moreover,

$$r_2^2 + r_3^2 = \frac{36}{1024}, \quad \frac{(3/32)(89/32)}{3} = \frac{89}{1024}, \quad \left(\frac{7}{32} \right)^2 = \frac{49}{1024}. \quad (9)$$

All inequalities in (2) therefore hold, and ρ_d is $A-BC$ PPT.

It is not $A-BC$ separable. The branch parameter (3) equals

$$u = \frac{1 - 1/32 - 2(3/4)}{1 - 3/32} = -\frac{17}{29}, \quad (10)$$

which lies strictly between -1 and 0 . But the two sides of (4) are respectively

$$3 \left(\frac{3}{16} \right)^2 + \left(1 + \frac{1}{32} - \frac{3}{4} \right)^2 = \frac{189}{1024}, \quad \left(2 - \frac{4}{32} - \frac{3}{2} \right)^2 = \frac{144}{1024}. \quad (11)$$

The necessary inequality fails strictly, by $45/1024$. Hence ρ_d is $A-BC$ entangled.

Finally, conjugation by $V_{(123)}$ fixes R_+ , R_- , R_0 , and R_3 , hence fixes ρ_d . This conjugation cyclically permutes the three tensor factors, and therefore carries the PPT cone and the separable cone for one cut onto those for the next cut. The $A-BC$ conclusions consequently hold for $B-AC$ and $C-AB$ as well. $\square$

5 Why the quoted obstruction does not apply

Remark 5.5 of [1] rules out an all-cuts example under any of three coefficient symmetries, each of which also requires $a_{(123)} = a_{(132)}$. In (5), the R_3 term contributes

$$\frac{9i}{32\sqrt{3}d(d^2-1)}(V_{(123)} - V_{(132)}),$$

so these two coefficients are unequal. This is precisely the transposition-symmetry-breaking direction invisible in the ruled-out slices.

6 Verification and novelty limitations

The included script `code/verify_candidate.py` checks (7)–(11) using exact rational arithmetic. It also constructs (5) directly for $d = 3, 4, 5$ and checks trace, positivity, coordinates, cyclic invariance, and all three partial transposes numerically. These matrix checks are safeguards; the proof above is dimension-independent and analytic.

A bounded exact-phrase and keyword search found the source question, Eggeling–Werner’s characterization, generic all-cuts PPT-entangled states, and broader tripartite-entanglement work, but no later explicit solution in the tripartite Werner class. This is not an exhaustive bibliographic proof. A specialist should check forward citations and databases before any public novelty claim.

References

- [1] S.-J. Park, Y.-G. Jung, J. Park, and S.-G. Youn, *A universal framework for entanglement detection under group symmetry*, arXiv:2301.03849v2 (2023), Remark 5.5.
- [2] T. Eggeling and R. F. Werner, *Separability properties of tripartite states with $U \otimes U \otimes U$ symmetry*, Phys. Rev. A 63 (2001), 042111; arXiv:quant-ph/0010096.